

Minu Maria Sabu

Aspiring Python Developer

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PROFESSIONAL SUMMARY

Electronics and Biomedical Engineering graduate with a strong foundation in biomedical instrumentation, medical devices, and programming. Currently building full-stack development skills through a Python Full Stack Development program. Experienced in hands-on hardware-software projects involving sensors, microcontrollers, and data-driven applications. Motivated to grow as a software professional and contribute to impactful, technology-driven solutions.

SKILLS

Technical: Python, SQL, HTML, CSS, C, Arduino/Embedded Systems

Professional: Communication, Problem-Solving, Collaboration, Management

PROJECTS

Gesture Identification Using Smart Glove — Academic Project

- Developed a smart glove for sign language recognition using flex sensors and Arduino Uno.
- Detected finger bending via voltage changes and compared sensor data to predefined threshold ranges to recognize hand gestures.
- Converted recognized gestures into text and then speech using an Android text-to-speech application to assist communication.

Stress Regulation Through HRV, BPV and Cardiac Coherence Biofeedback — Academic Project

- Built a stress management and mental well-being system using HRV coherence biofeedback and BPV analysis.
- Analyzed stress levels from ECG and PPG signals acquired via Arduino Uno and a ProtoCentral MAX86150 PPG/ECG breakout board.
- Calculated coherence scores reflecting heart-brain balance and integrated guided breathing exercises with animated visualizations.
- Added blood pressure analysis, wellness activities, and an interactive reward-based UI to boost user engagement (Arduino IDE, VS Code, Next.js).

WORK EXPERIENCE

Biomedical Trainee — Sunrise Hospital, Ernakulam Apr 2025 – Jul 2025

Biomedical Intern — Cyrix Healthcare Private Limited, Ernakulam Jul 2024 – Jan 2025

Biomedical Student Intern — Welcare Hospital, Ernakulam May 2023

- Visited hospital departments to learn the function and operation of medical equipment.
- Assisted with departmental rounds and attended a demonstration program for new equipment.

EDUCATION

B.Tech, Electronics and Biomedical Engineering — Govt. Model Engineering College 2020 – 2024

Relevant coursework: Biomedical Instrumentation, Medical Devices, Medical Imaging, Biomedical Signal Processing, Electronic Circuits, Digital Electronics, Microprocessors and Microcontrollers, Sensors and Transducers, Programming Fundamentals (C & Python).

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL Online Certification (IIT Guwahati) — Human Behaviour, Elite, 84% (Feb–Apr 2024)
- Certificate of Participation — Cloud Computing Workshop, Excel 2021, Govt. Model Engineering College

- Certificate of Participation — National Event "Electric and Future Mobility," Mar Athanasius College of Engineering (May 2021)
- Technopreneur 2021 National Entrepreneurial Symposium — Speaker Sessions; Workshops on Product Management, Internet of Things, and UX Driven UI Perspectives